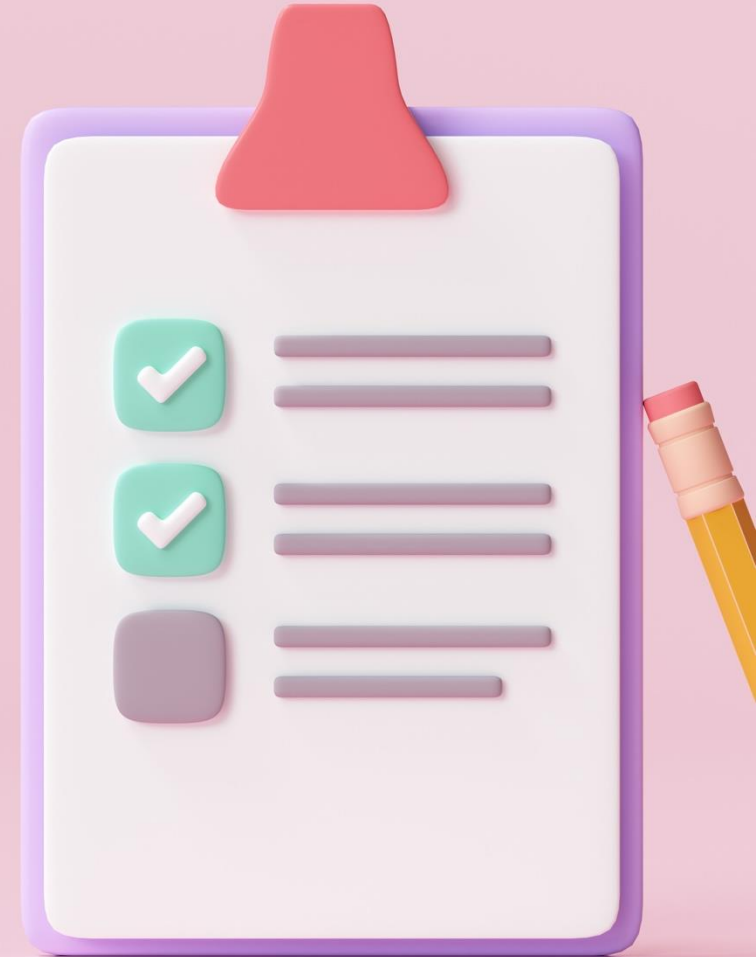


Sprint 4:

Inference and confidence



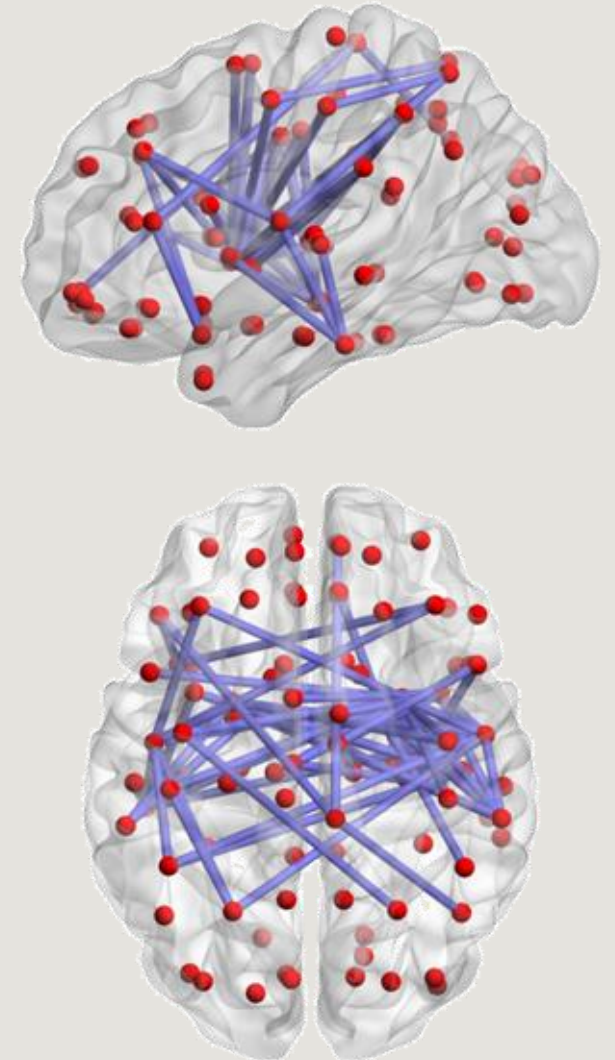
Agenda

Statistical Inference

Margin of Error

Confidence Intervals

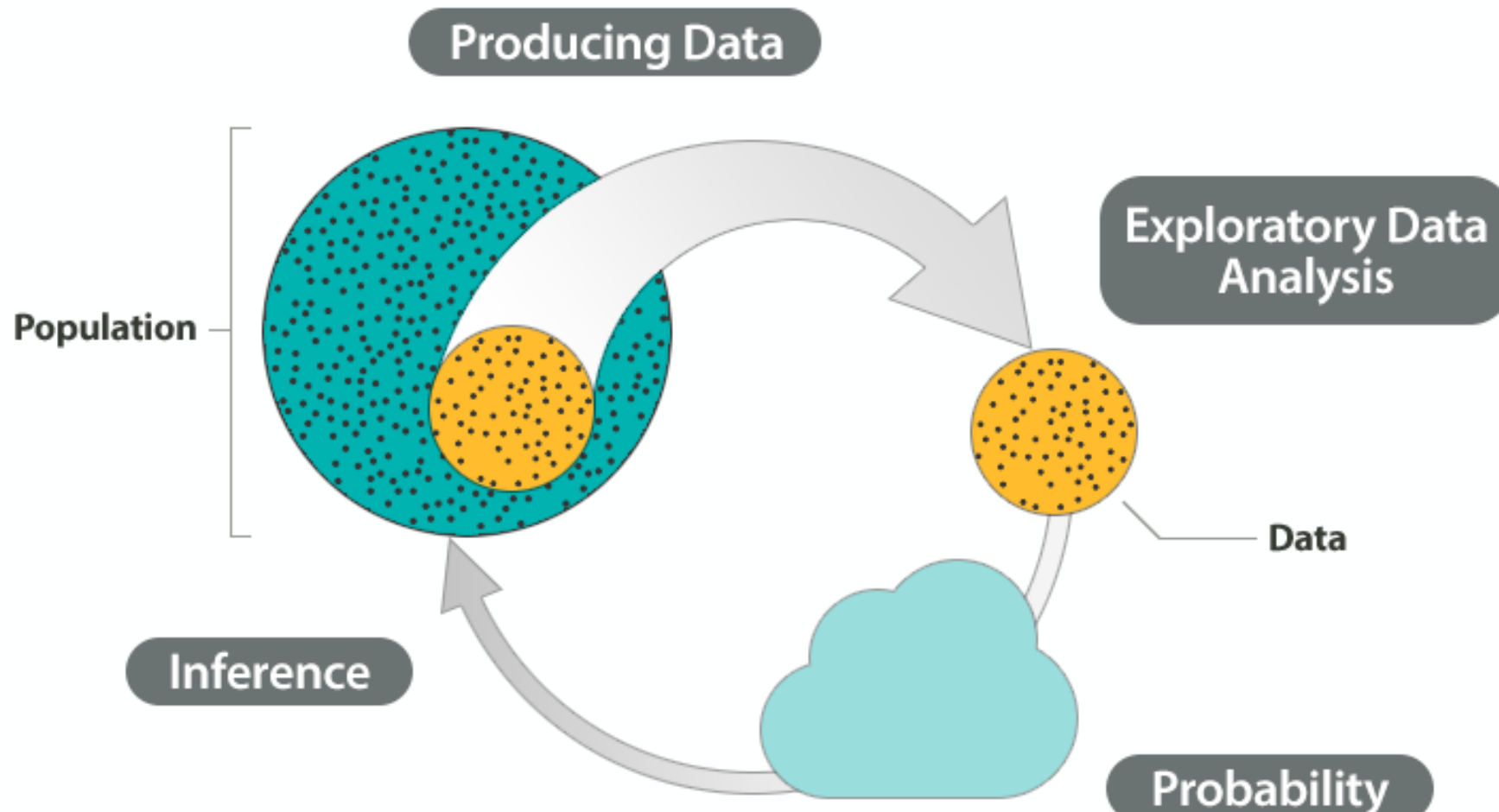
T-distribution



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Statistical Inference

- **Inference:** drawing conclusions about our population based on what we've discovered about our sample.

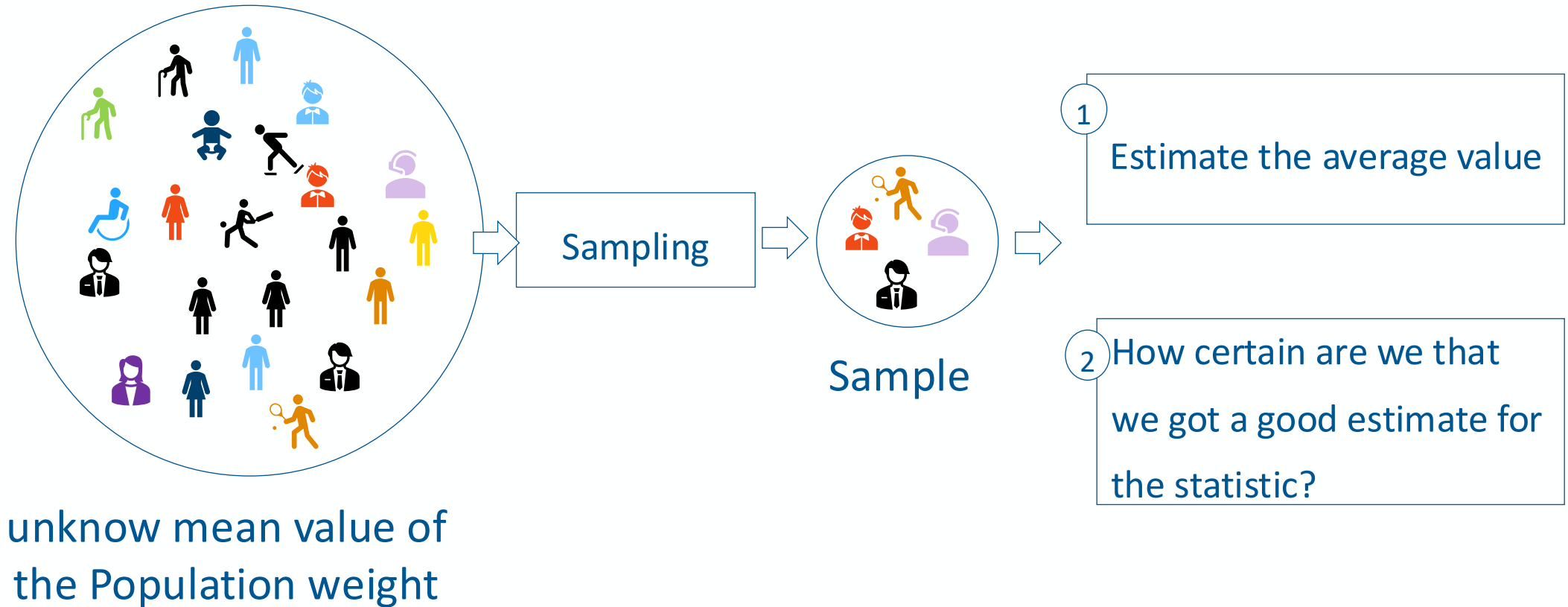


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We first estimate then we generalize

- The true parameters of population distribution are unknown; we need to estimate them from the **sample**.
- **Estimation** is finding the most likely values of population statistics such as the mean μ or proportion p
- For this, we need an **estimator which** is a rule or formula that uses **sample data** to estimate an unknown **population parameter** (like a mean μ , proportion p or variance).

Example: we first estimate then we generalize

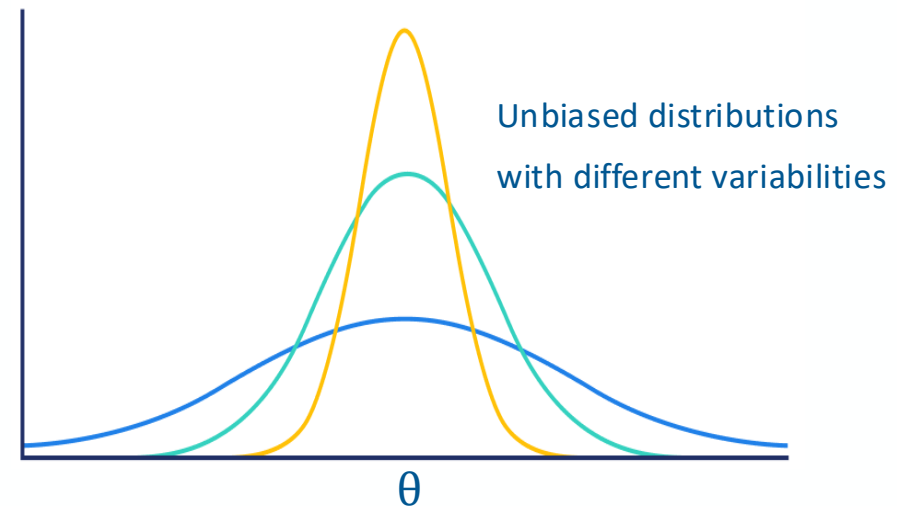
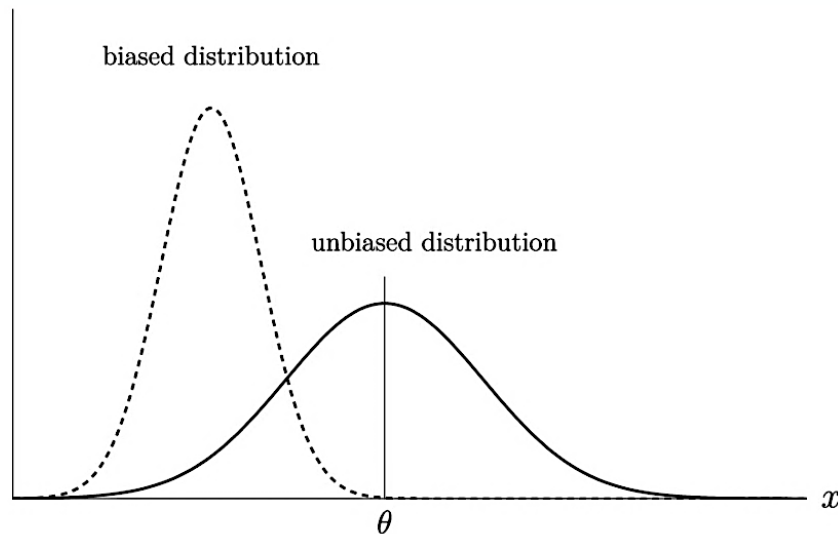


We first estimate then we generalize

- Unbiased estimator does **not systematically overestimate or underestimate** the parameter.
- An estimator is **unbiased** if, **on average over many samples**, it equals the true population parameter.

$$\theta \approx E[\hat{\theta}] \rightarrow \text{Our best estimate of the true population parameter}$$

- Of all the unbiased estimators, we prefer the estimator whose **sampling distribution has the smallest spread or variability (n is high)**
- The shape of sampling distributions depend on sample sizes. **E.g. To get the smallest variability, the sample size should be high.**



Estimation of population mean

Example Unbiased estimation of population mean μ is Sample mean $\bar{x}$

$$\bar{x} = \frac{1}{n} \sum_{i=0}^n x_i$$

Over many repeated samples, the average of all sample means equals μ

Estimation of population standard deviation

Unbiased estimation of unknown population standard deviation σ is sample standard deviation s

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

- **What each symbol means**
 - x_i : individual sample observations
 - $\bar{x}$ sample mean
 - n : sample size
 - s : **estimate we go with for the population standard deviation σ**

Estimation of population proportion

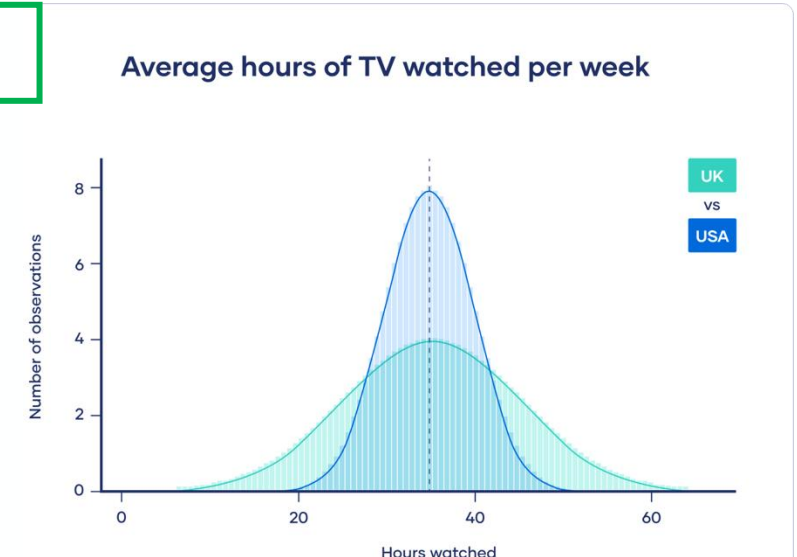
Unbiased estimation of unknown population proportion p is sample proportion $\hat{p}$

$$\hat{p} = \frac{x}{n}$$

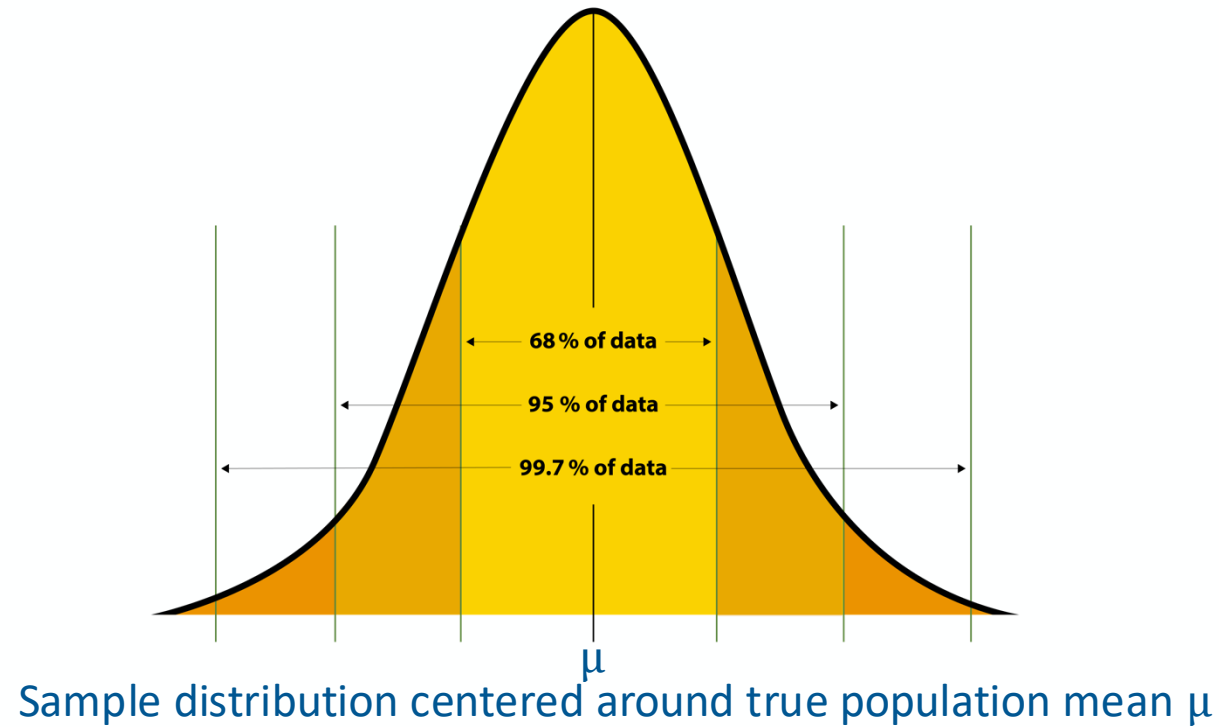
- **What each symbol means**
 - x : number of successes in the sample (e.g. students who are higher than 180 cm)

Quiz

- Researchers estimate the **average number of hours spent watching TV per week** in the United Kingdom and the United States. Individual TV-watching habits are **more variable in the UK than in the US**.
- **Assume** both countries have the same true average weekly TV-watching time.
- **Which statement is correct?**
 - A. The UK sample mean will be systematically higher than the US sample mean
 - B. Higher variability in the UK makes the estimated mean less precise
 - C. Variability does not affect estimation of the mean
 - D. Higher variability guarantees bias in the estimate



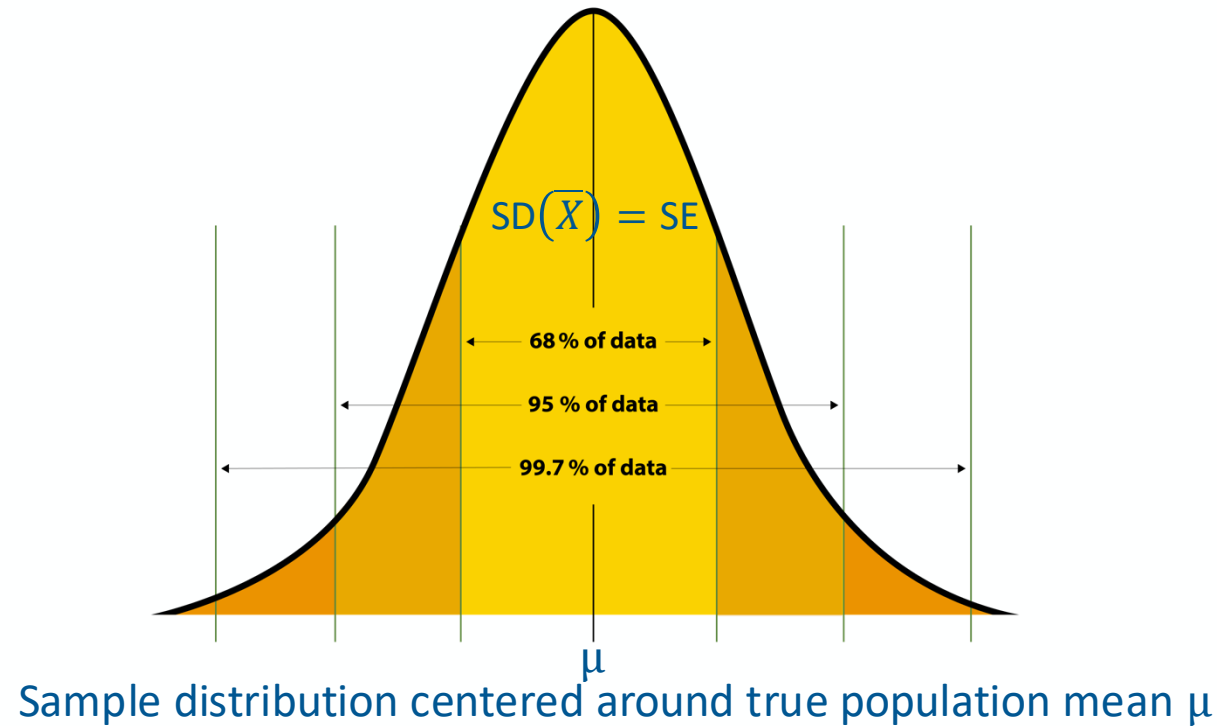
Certainty in our estimate of mean value



Single sample mean or $\hat{\mu}$ is an estimator of true population mean

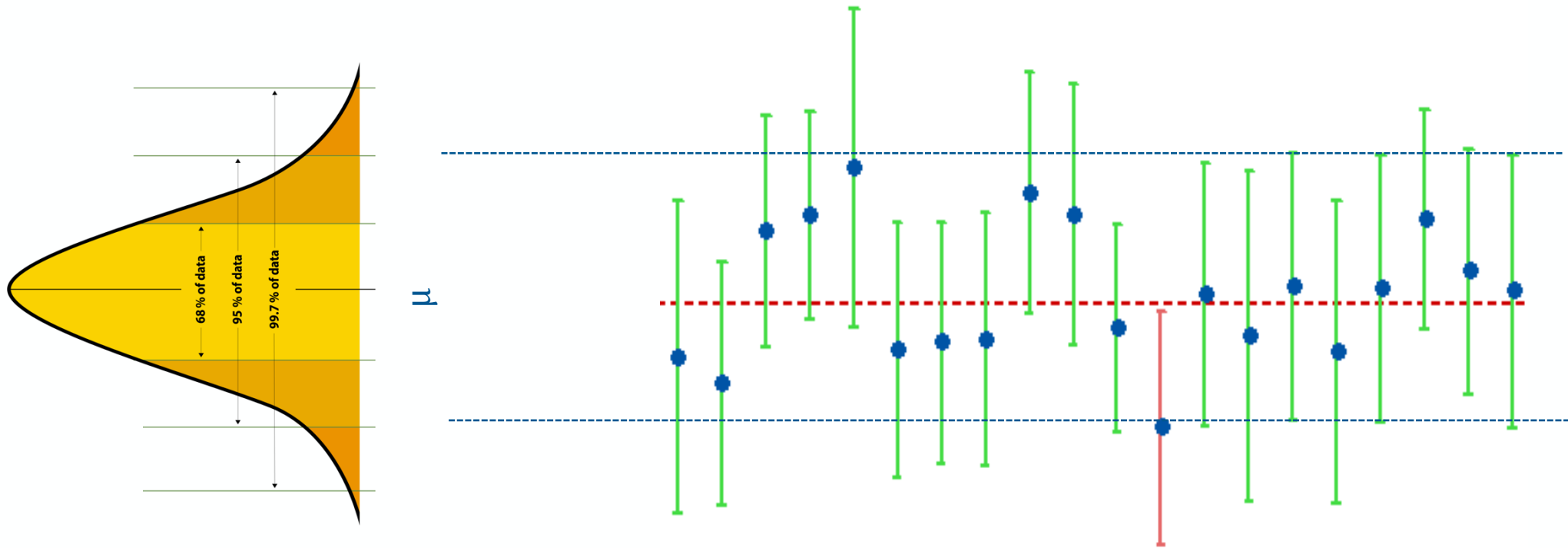
$$\mu \approx E[\hat{\mu}]$$

Certainty in our estimate of mean value



If we want to capture **exactly 95%** of the data, we need to find a threshold to scale standard deviation of the sample distribution which is called standard error:

$$SE = SD(\bar{X}) = \frac{\sigma}{\sqrt{n}}; \sigma \text{ is population standard deviation}$$

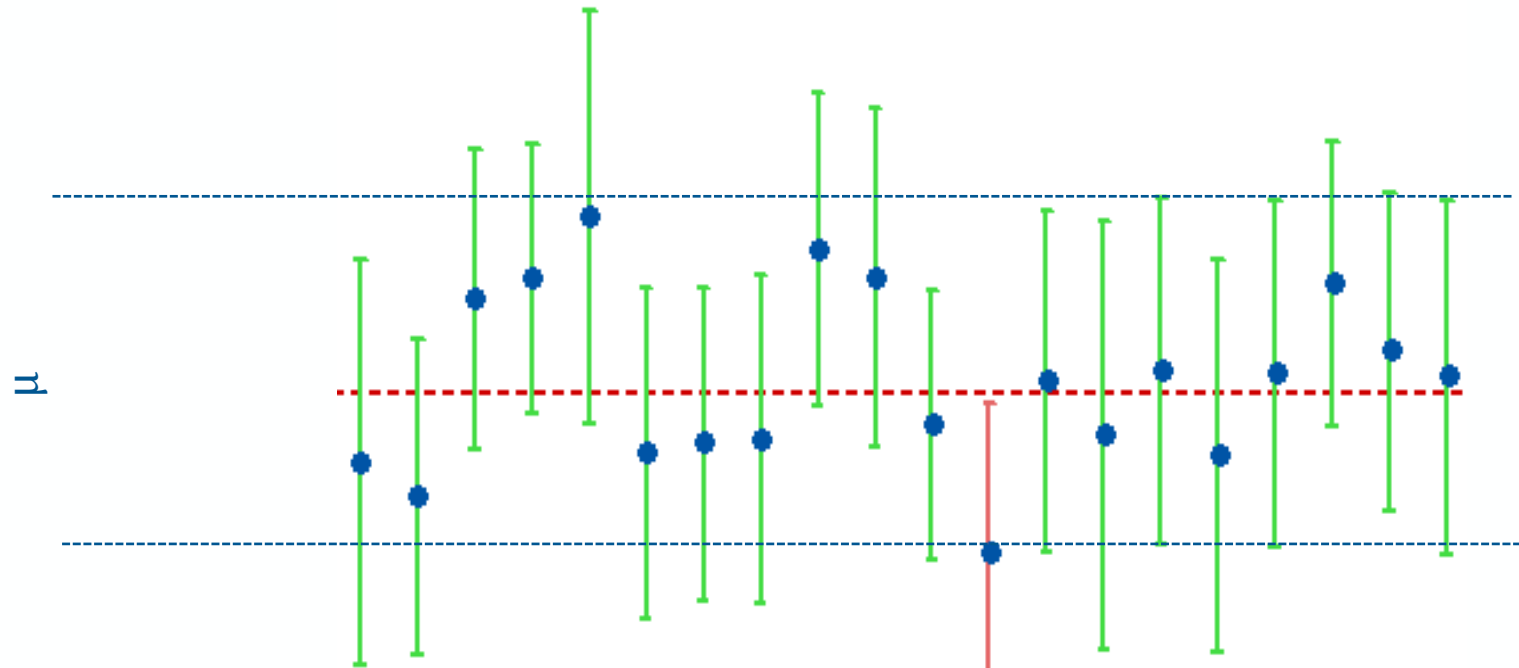
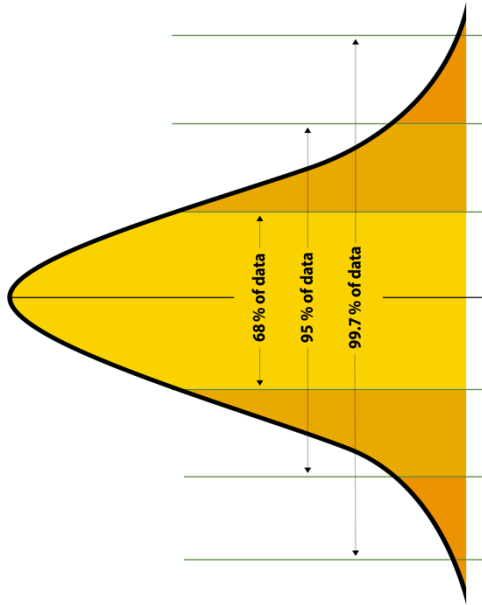


About 95% of sample means will lie within $\pm c_{\alpha/2} \times SE$ of the true population mean. *

Equivalently: If we repeatedly draw samples of the same size and compute a confidence interval each time, **about 95% of those intervals will contain the true population mean.**

- (left) Normalized (standardized) distribution of sample means. The standard deviation of this distribution is the standard error
- (Right) individual sample means in blue dots and individual intervals of 95% of the data points in green and red *

We can use the 1.96 from Z table as critical value $c_{\alpha/2}$ for the 95% confidence when sample size relatively large ($n \geq 30$) or the σ of population is known



About 95% of sample means will lie within $\pm c_{\alpha/2} \times SE$ of the true population mean.*

Equivalently: If we repeatedly draw samples of the same size and compute a confidence interval each time, **about 95% of those intervals will contain the true population mean.**

$$P(\mu \in [\hat{\mu} \pm c_{\alpha/2} \times SE]) \approx 0.95$$

Margin of error if we want that 95% of all sample means to be within $\pm c_{\alpha/2} \times SE$ around the true population mean.

Level of confidence in our estimate

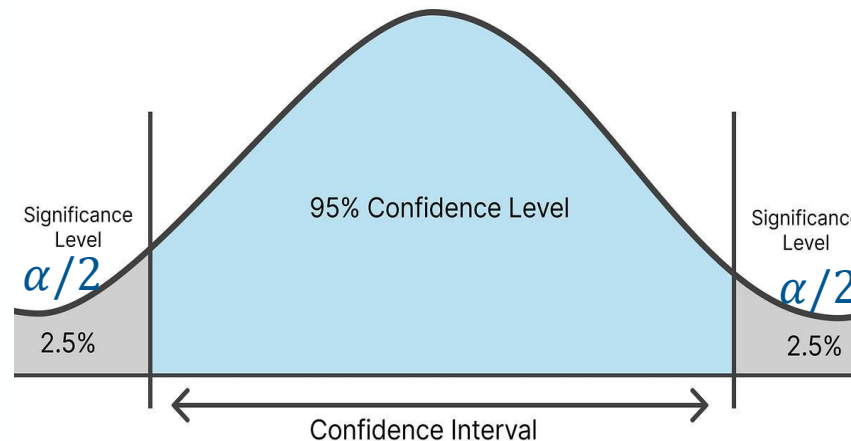
We can use the 1.96 from Z table as critical value $c_{\alpha/2}$ for the 95% confidence when sample size relatively large ($n \geq 30$) or the σ of population is known

Analogy

- If I repeatedly take 100 samples of 35 students age each, about 95% of the sample means will fall within $\mu \pm 1.96 \times$ **standard errors** of the true population mean.
- If I repeatedly take samples of 35 students age, about **90%** of the sample means will fall within $\mu \pm 1.64 \times$ **standard errors** of the true population mean.
- If I repeatedly take samples of 35 students age, about **99%** of the sample means will fall within $\mu \pm 2.58 \times$ **standard errors** of the true population mean.
- Increasing **X%** makes the interval wider; decreasing it makes the interval narrower.
- It is a range around the sample mean that is likely to contain the true population mean, with a specified level of confidence (such as 95%).

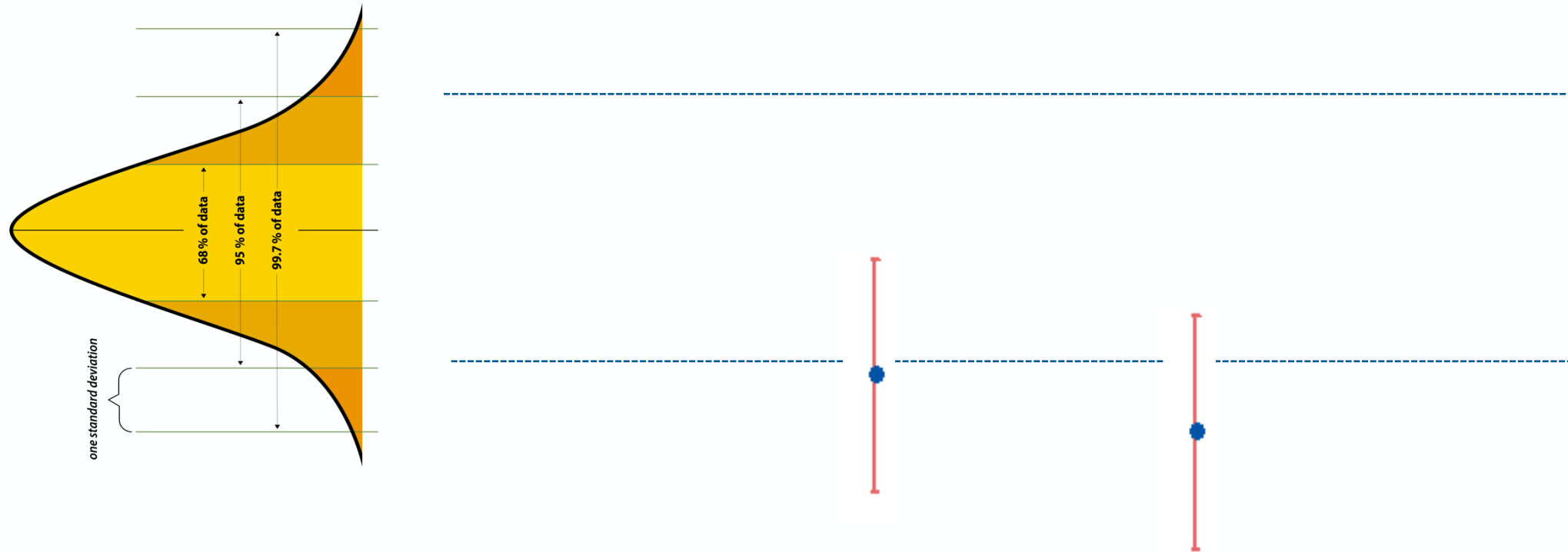
Margin of error

- **Margin of error** is a statistical term that represents the range of uncertainty surrounding an estimate.
- It reflects how much the estimate is expected to vary from the true population value and indicates the **level of confidence** we have in the estimate's accuracy.
- **Three factors affect the size of MOE:**
 - Confidence level ($1-\alpha$): up to us to decide, where α is 0.05 if we want confidence level of 95%
 - The sample size: increasing it will decrease MOE
 - The amount of variability in the population: Less variability means less MOE



Quiz

- We took two samples at different time points, when would you consider the red interval a result of random sampling?



- Sample Variability:** The red interval can result from natural fluctuations in random sampling, especially with small samples. But if its mean is far from the population mean, it may suggest issues with the reported population mean or other sources of variation.

how to choose confidence level and the corresponding critical value $c_{\alpha/2}$?

- The **confidence level** corresponds to the percentage of the time; our estimates would be correct if we take numerous random samples. Typical confidence levels would be 95% or 99%.
- The confidence level choice determines the critical value the we use to scale the Standard error SE we add or subtract to get the confidence level we desire.

how to choose confidence level and the corresponding critical value $c_{\alpha/2}$?

Example **95% confidence**:

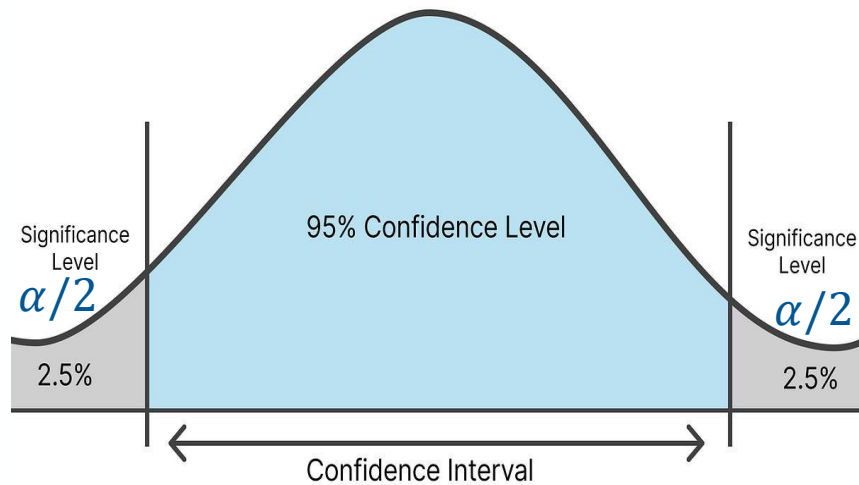
1. If population standard deviation is known:

We can use **z table** to find $c_{\alpha/2}$ and label it as z^* or $z_{\alpha/2}$

2. If population standard deviation is unknown:

- Sample size is small ($n < 30$): we can rely on the sample standard deviation, but we need another approximation to approximate the area the critical value (using Student's t-distribution) and the critical value is denoted as t^* with proper degree of freedom or $t_{\alpha/2, d.f}$ (will be addressed later)
- Sample size is large ($n \geq 30$): the result from t-distribution will be approximately the same as the result we would get from the z table z^* or $z_{\alpha/2}$

1. Margin of error if population standard deviation is known



Tail area $\alpha/2$	$z_{\alpha/2}$
.05	1.645
.025	1.96
.005	2.58

- Using z distribution to define the upper and lower limit for 95% confidence applies only for $n > 30$ or when the CLT applies. In that case, we find $z_{\alpha/2}$ from the Z table

2. Margin of error if population standard deviation is unknown

- If σ is the true population standard deviation and it is usually unknown; we can rely to past studies or conduct a pilot study and use the sample standard deviation s as a substitute for σ . That estimate is then **plugged into the formula**:

$$SE = \frac{s}{\sqrt{n}}$$

- Our best estimate of standard error is the standard error of the estimator

$$\hat{SE} = \frac{s}{\sqrt{n}}$$

In this case:

- Sample size is small ($n < 30$): we can rely on the sample standard deviation, but we need another approximation to approximate the area the critical value (using Student's t-distribution) and the critical value is denoted as t^* with proper degree of freedom or $t_{\alpha/2, d.f}$ (will be addressed later)
- Sample size is large ($n \geq 30$): the result from t-distribution will be approximately the same like the result we would get from the z table z^* or $z_{\alpha/2}$

Factoring in sample size

- When conditions are met for normal approximation, we write:

$$\text{MOE} = \pm c_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

The larger n the less the MOE is.

- If we must determine sample size to achieve a certain MOE, we can use

$$n = \left\lceil \left(\frac{c_{\alpha/2} \times \sigma}{\text{MOE}} \right)^2 \right\rceil$$

(the absence of ledges on the bottom means you are supposed to round up your result to the next greatest integer)

Quiz

- Researchers want the **same level of precision** when estimating the mean TV-watching time in UK and US.
- **What should they do?**
 - A. Use a larger sample in the US
 - B. Use a larger sample in the UK
 - C. Use identical sample sizes
 - D. Reduce variability in the UK population

Example

- A **German census Bureau** wants to estimate the **average annual 3-person household electricity consumption** in a large state. From previous census studies, the population consumption standard deviation is estimated to be ($\sigma = 2500$) kWh.
- The Census Bureau wants the estimate of the population mean to be **within 2% of the true mean**, which is approximately **5000 kWh**, with **95% confidence**.
- **Answer:** known population standard deviation with confidence level: 95% $\Rightarrow (c_{\alpha/2} = z_{\alpha/2} = 1.96)$

$$n = \left\lceil \left(\frac{z_{\alpha/2} \times \sigma}{\text{MOE}} \right)^2 \right\rceil = \left\lceil \left(\frac{1.96 \times 2500}{0.02 \times 5000} \right)^2 \right\rceil = 2401$$

* usually variability is more stable, that is why using old population standard deviation works well here

MOE of proportion estimate

- Standard error for an estimate of population proportion p is

$$SE(\hat{p}) = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

- where $\hat{p}$ is sample proportion

$$\hat{p} = \frac{x}{n}$$

- If $np > 10$ and $nq > 10$, then we can use Z table to find the critical value $z_{\alpha/2}$:

$$\text{MOE: } \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

- Notice:** we don't use t-distribution to find the critical value for MOE of proportion in any case.

Population size for certain MOE of proportion estimate

$$\text{MOE: } \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

$$n = \left\lceil \hat{p}(1 - \hat{p}) \left(\frac{z_{\alpha/2}}{\text{MOE}} \right)^2 \right\rceil$$

- When we don't know anything about $\hat{p}$, we can **use ($\hat{p} = 0.50$)** because it is the **most conservative choice**.
- Here's why, briefly:
 - The margin of error for a proportion depends on $(\hat{p}(1 - \hat{p}))$ and this quantity $(\hat{p}(1 - \hat{p}))$ is **maximized at ($\hat{p} = 0.50$)**.
 - This produces the **largest possible variance** and therefore the **largest required sample size**.
 - Using (50%) guarantees the sample size will be sufficient **no matter what the true proportion is**.
 - So, when no prior estimate of the proportion is available, assuming **($\hat{p} = 0.50$)** is a safe and standard practice.

Quiz

- The **U.S. Census Bureau** wants to estimate the **proportion of households with internet access** in a large state.
- To ensure accuracy, the estimate must be **within 2 percentage points (0.02)** of the true population proportion, with **95% confidence**. Since the true proportion is unknown, the Census Bureau uses a **conservative estimate**.
- What is the **minimum sample size (n)** required?
- **Given**
 - Margin of error: (MOE = 0.02)
 - Confidence level: 95% (z = 1.96)
 - Use ($\hat{p} = 0.5$)

$$n = \left\lceil \hat{p}(1 - \hat{p}) \left(\frac{z_{\alpha/2}}{\text{MOE}} \right)^2 \right\rceil = \left\lceil (0.5)(0.5) \frac{(1.96)^2}{(0.02)^2} \right\rceil = 2401$$

<https://select-statistics.co.uk/calculators/sample-size-calculator-population-proportion/>

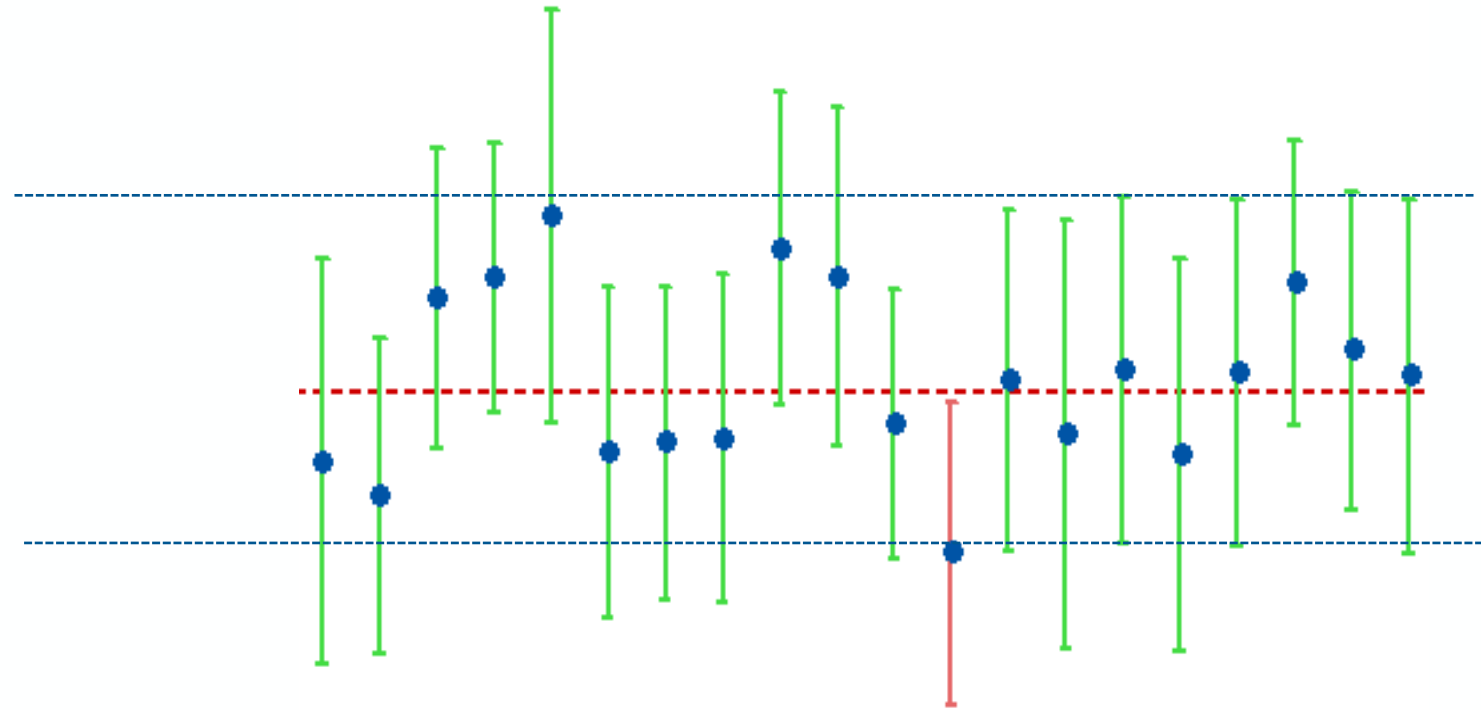
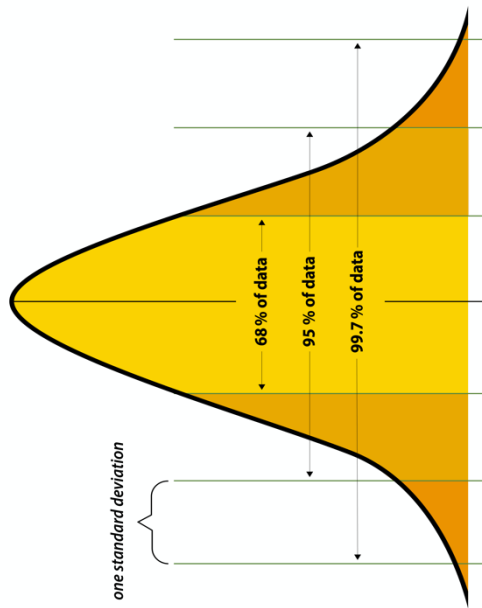
Confidence Interval for Quantitative Population

- When a characteristic being measured such as IQ, price, income, weight...etc is numerical people often an estimate of the mean of the population.
- To do so, we use the sample mean as an estimation plus or minus the margin of error:

$$\bar{X} \pm c_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

- Important: we use her the population standard deviation (otherwise, we must rely on t-distribution)
- We call this confidence interval where we are sure (to some degree) that the parameter lies between these two values.

Important: Interpretation of confidence interval



About 95% of sample means will lie within $\pm c_{\alpha/2} \times SE$ of the true population mean.

Equivalently: If we repeatedly draw samples of the same size and compute a confidence interval each time, **about 95% of those intervals will contain the true population mean.**

- Only if the estimator falls in the tail areas will the interval fail to enclose the parameter. This happens only 5% of the time.

Interval Estimation/Confidence Interval

- When we estimate an important parameter from our sample, we can create an interval (a, b) so that you are fairly sure that the parameter lies between these two values.
- “Fairly sure” in this context means “with quantifiable relatively high probability”, measured using the **confidence coefficient, $1-\alpha$** .

Usually, $1-\alpha = .90, .95, .99$ and $100(1-\alpha)\%$ Confidence Interval for the point estimator is

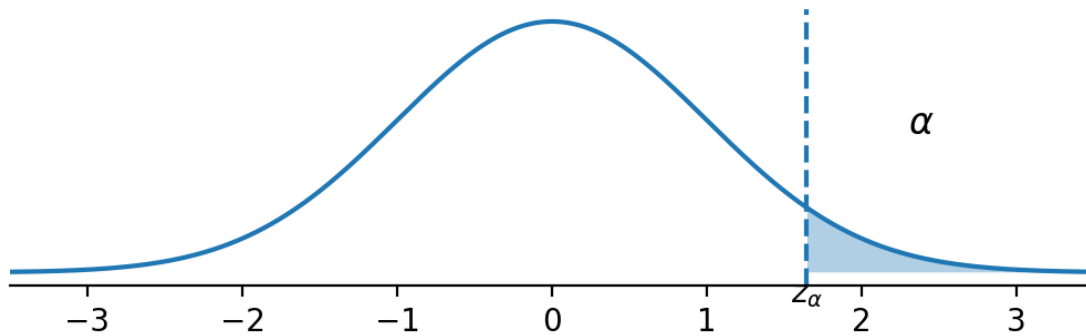
$$\text{Point Estimator} \pm c_{\alpha/2} \text{SE}$$

For large-Sample size or when population variance is known, $c_{\alpha/2}$ could be found from z table, and the Confidence Interval is:

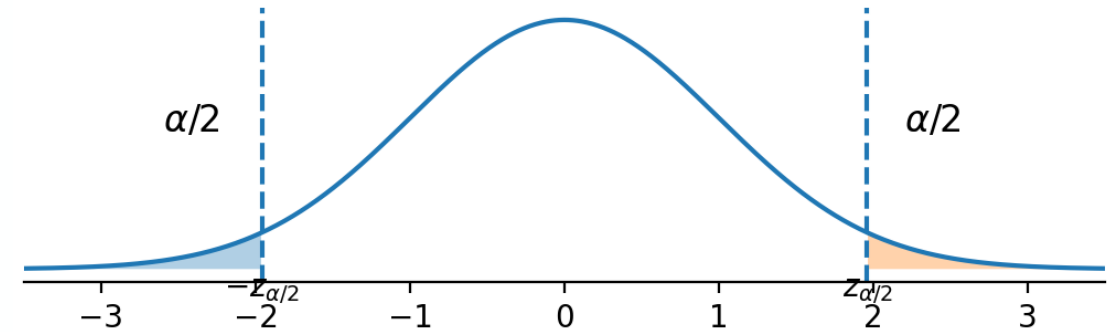
$$\text{Point Estimator} \pm z_{\alpha/2} \text{SE}$$

Attention (out of context): One-Tailed vs Two-Tailed Statistical Tables

One-tailed (upper tail): $P(Z > z_\alpha) = \alpha$



Two-tailed: $P(|Z| > z_{\alpha/2}) = \alpha$



One-tailed table: $\alpha = 0.05 \rightarrow z_\alpha = z_{0.5} = 1.645$

Two-tailed table: $\alpha = 0.05 \rightarrow z_{\alpha/2} = z_{0.025} = 1.96$

Example

- A high-integrity systems monitoring team analyzes long-term operational data. Historical records show that the **daily system performance metric has a population standard deviation of 35 units.**
- A random sample of **n = 50** system logs yields a **mean performance value of 756 units.** Find the 95% confidence interval.

Solution:

- Confidence level is $(1 - \alpha) = 0.95$.
- Population standard deviation is known=> we can use z table to find $z_{\alpha/2} = 1.95$

$$\bar{x} \pm z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) = 756 \pm 1.96 \left(\frac{35}{\sqrt{50}} \right) = 756 \pm 9.70$$

So, the confidence interval is (746.30, 765.70)

We are 95% confident that the true mean system performance lies between **746.3 and 765.7 units.**

This means that if we repeatedly sampled and built confidence intervals in the same way, **95% of those intervals would contain the true mean.**

Example

- A high-integrity systems monitoring team analyzes long-term operational data. Historical records show that the **daily system performance metric has a population standard deviation of 35 units.**
- A random sample of **n = 50** system logs yields a **mean performance value of 756 units.** Find the 99% confidence interval.

Solution:

- Confidence level is $(1 - \alpha) = 0.99$.
- Population standard deviation is known, and the number of units is greater than 30 $\Rightarrow$ we can use z table to find $z_{\alpha/2} = 2.58$

$$\bar{x} \pm z_{\alpha/2} \left(\frac{\sigma}{\sqrt{n}} \right) = 756 \pm 2.58 \left(\frac{35}{\sqrt{50}} \right) = 756 \pm 9.70$$

So, the confidence interval is (743.24, 768.76)

Increasing confidence level increased the confidence interval. The interval must be wider to provide for the increased confidence that it does indeed enclose the true value of μ .

Confidence Interval For a binomial population

- When a proportion being measured such as percentage of people over pension age or proportion of people who would vote for a specific political party, people often asks for a confidence interval for their estimate
- To do so, we use the sample proportion as an estimation plus or minus the margin of error of the proportion

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

Important: $\hat{p}$ is sample proportion

- Unlike means, where we use the T distribution if the population standard deviation is unknown, proportions **always use Z**. This is because the standard error of a proportion is calculated directly from the sample proportion itself, rather than needing an independent estimate of the population's standard deviation

Example

- Of a random sample of $n = 50$ HIS students, 44 of the students said that they find IDA course great. Estimate the proportion of total HIS students who find IDA great course with 90% confidence interval.

Solution:

- Confidence level is $(1 - \alpha) = 0.90$.
- Population standard deviation is known, and the number of units is greater than 30 $\Rightarrow$ we can use z table to find $z_{\alpha/2} = 1.645$

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = 0.88 \pm 1.645 \sqrt{\frac{0.88(1 - 0.88)}{50}} = 0.88 \pm 0.076$$

So, the confidence interval $\approx (0.8, 0.95)$

We are 90% confident that the true population proportion lies between 0.8, 0.95

Estimating the Difference between Two Means

- Sometimes we are interested in comparing the means of two populations.
 - The average failure rates of systems with two different maintenance strategies.
 - The average scores for students taught with two different teaching methods.
- To make this comparison:
- We need two different samples n_1 and n_2 , with two different standard deviations and mean values: $\mu_1, \mu_2, \sigma_1, \sigma_2$

	Mean	Variance	Standard Deviation
Population 1	μ_1	σ_1^2	σ_1
Population 2	μ_2	σ_2^2	σ_2



	Sample size	Mean	Variance	Standard Deviation
Sample from Population 1	n_1	$\bar{x}_1$	s_1^2	s_1
Sample from Population 2	n_2	$\bar{x}_2$	s_2^2	s_2

Estimating the Difference between Two Means

- The best estimate of $\mu_1 - \mu_2$ is the difference in the two sample proportions $\bar{x}_1 - \bar{x}_2$
- The confidence interval is computed using:

$$\bar{x}_1 - \bar{x}_2 \pm c_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

- Finding critical value:
 - If standard deviations of populations are known, we use Z table to determine the critical value
 - If standard deviations of populations are unknown:
 - To determine the critical value, we must rely on t-distribution with $n_1 + n_2 - 2$ degrees of freedom
 - If both $n_1 > 30$ and $n_2 > 30$, the t-distribution closely approximates the standard normal, so using the Z-table is acceptable.

Example

- Compare the average score of two subjects, IDA and Data Mining using a 95% confidence interval

Avg score	IDA	Data Mining
Sample size	50	50
Sample mean	68	59
Sample Std Dev	35	30

- The population std is unknown but since both sample sizes >30, we can rely on z table to find approximate critical value for 95% confidence. Important the **correct distribution is still the t distribution** in theory, because σ_1 and σ_2 are unknown.

$$\bar{x}_1 - \bar{x}_2 \pm_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

$$68 - 59 \pm 1.96 \sqrt{\frac{35}{50} + \frac{30}{50}}$$

We are 95% confident that the true difference in population means lies between **6.77 and 11.23**

Estimating the Difference between Two proportions

- The best estimate of $m_1 - m_2$ is the difference in the two sample means $\bar{x}_1 - \bar{x}_2$
- The confidence interval is computed using:

$$\bar{x}_1 - \bar{x}_2 \pm c_{\alpha/2} \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

- Finding critical value:
 - If standard deviations of populations are known, we use Z table to determine the critical value
 - If standard deviations of populations are unknown:
 - To determine the critical value, we must rely on t-distribution with $n_1 + n_2 - 2$ degrees of freedom
 - If both $n_1 > 30$ and $n_2 > 30$, we can use Z table as an approximation to find critical value, but the right distribution is t with $n_1 + n_2 - 2$ degrees of freedom

Estimating the Difference between Two Means

- The best estimate of $P_1 - P_2$ is the difference in the two sample proportions $\hat{p}_1 - \hat{p}_2$
- The confidence interval is computed using:

$$\hat{p}_1 - \hat{p}_2 \pm z_{\alpha/2} \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

Example: Difference in Two Population Proportions (Election Example)

- In the **2020 U.S. presidential election**, a post-election survey compares **Democratic** support between urban and rural voters.
- **Urban voters:** $n_1 = 2400$, $x_1 = 1440$ voted Democratic $\hat{p}_1$
- **Rural voters:** $n_2 = 2200$, $x_2 = 1100$ voted Democratic $\hat{p}_2$

Task: Construct a **95% confidence interval** for $p_1 - p_2$ and interpret the results

$$\hat{p}_1 = \frac{1440}{2400} = 0.60, \quad \hat{p}_2 = \frac{1100}{2200} = 0.50$$

$$SE = \sqrt{\frac{0.60(0.40)}{2400} + \frac{0.50(0.50)}{2200}} \approx 0.01462$$

$$MOE = 1.96(0.01462) \approx 0.02865$$

Confidence interval for 95% confidence is 0.10 ± 0.02865 ; = (0.07135, ; 0.12865)

Interpretation: We're 95% confident the urban Democratic vote proportion is about **7.1 to 12.9 percentage points higher** than the rural proportion.

The Problem (σ Unknown)

- If population standard deviation σ is **known**

$$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$$

- In practice, σ is **unknown** → we must replace with sample SD s :

$$\frac{\bar{x} - \mu}{s/\sqrt{n}}$$

- Since s is **not fixed among samples**, the statistic has **extra variability**
- Normal (z) distribution **underestimates uncertainty and we must pay extra penalty for this variability.**

Solution: t-distribution

- When σ is unknown or even if it is known and the sample size is small and we cannot be sure the population is normal then:

$$\frac{\bar{x} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

- **t-distribution:**
 - Heavier tails than normal
 - Accounts for uncertainty in estimating σ
 - Depends on **degrees of freedom = $n - 1$**
 - As $n \rightarrow \infty : t_{n-1} \rightarrow N(0,1)$

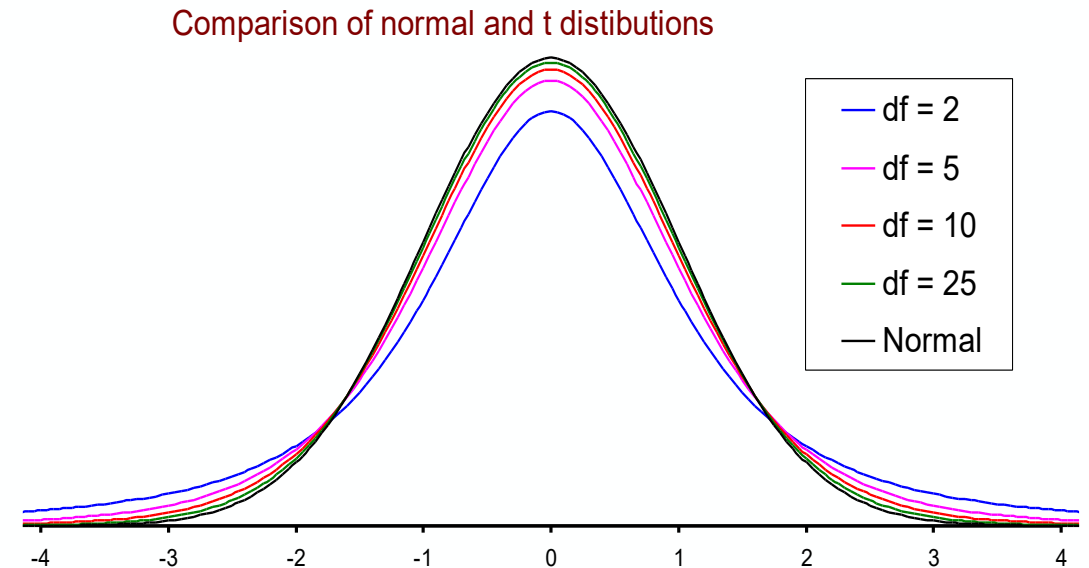
T-distribution vs. Normal Distribution

- **T-distribution is a cousin of the standard normal distribution:**

- Follows mathematical function
- Symmetrical, continuous, bell-shaped
- Continues to $\pm$ infinity
- Mean: $M=0$
- Area under curve = $p(\text{event}[s])$
- When N is **large** (≈ 300), $t \approx z$

- **However:**

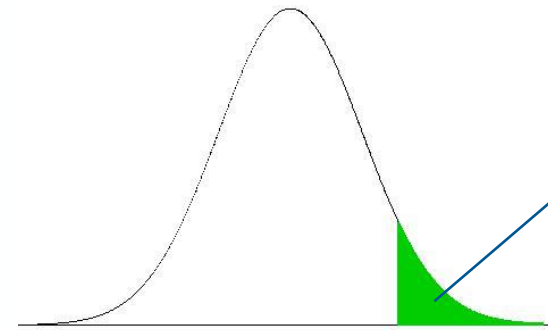
- more variable than z , with “heavier tails”
- Family of distributions
- $t^* > z^*$, for a given α



Using the t-Table

- Table 4 gives the values of t that cut off certain critical values in the right tail of the t distribution.
- Use index df and the appropriate tail area α to find t_{α} , the value of t with area α to its right.

df	$t_{.100}$	$t_{.050}$	$t_{.025}$	$t_{.010}$
1	3.078	6.314	12.706	31.821
2	1.886	2.920	4.303	6.965
3	1.638	2.353	3.182	4.541
4	1.533	2.132	2.776	3.747
5	1.476	2.015	2.571	3.365
6	1.440	1.943	2.447	3.143
7	1.415	1.895	2.365	2.998
8	1.397	1.860	2.306	2.896
9	1.383	1.833	2.262	2.821
10	1.372	1.812	2.228	2.764
11	1.363	1.796	2.201	2.718
12	1.356	1.782	2.179	2.681
13	1.350	1.771	2.160	2.650
14	1.345	1.761	2.145	2.624
15	1.341	1.753	2.131	2.602



α significance level

Critical value t_{α}

For a random sample of size $n = 10$, find a value of t that cuts off .025 in the right tail.

Row = $df = n - 1 = 9$

Column subscript = $\alpha = .025$

$t_{.025} = 2.262$

z vs t Distribution — Formulas Summary

Case	Use z or t	Test Statistic (LaTeX)	Degrees of Freedom
One mean (σ known)	z	$z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$	—
One mean (σ unknown)	t	$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$	$n - 1$
Two means (σ_1, σ_2 known)	z	$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$	—
Two means (unknown σ_1, σ_2 but equal var. of the samples)	t or Welch's t	$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$	$df = n_1 + n_2 - 2$
Two means (σ unknown, or unequal var.)	Welch's t	$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$	$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{\left(\frac{s_1^2}{n_1}\right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2}\right)^2}{n_2 - 1}}$

z vs t Distribution — Formulas Summary

Case	Use z or t	Test Statistic (LaTeX)	Degrees of Freedom
One proportion	z	$z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$	—
Two proportions	z	$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}}$	—

The most important skill is knowing **which test to use and how to apply it**

Example

Ten randomly selected students were each asked to report how many hours they spend on social media per month. The results are [82, 66, 90, 84, 75, 88, 80, 94, 110, 91].

Find a 90% confidence interval for the true mean number of social media hours per month by students.

$$n = 10, \bar{x} = 86.0, s = 11.84$$

- Since σ is unknown and n is small, use a **t-interval** with 90% **t-interval** with:

$$df = n - 1 = 9: t^* = 1.833$$

- Margin of error:

$$1.833 \cdot \frac{11.84}{\sqrt{10}} \approx 6.86$$

- 90% confidence interval:

$$(86 - 6.86, 86 + 6.86) = \boxed{(79.1, 92.9)}$$

Example

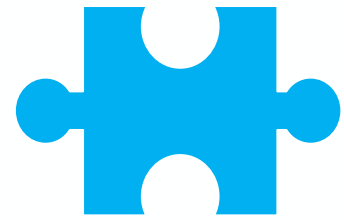
- An employee recorded the mileage he obtained while commuting to work with car. He kept track of the mileage for twelve different tanks of fuel, involving gasoline of two different octane ratings. Compute the 95% confidence interval for the difference of mean mileages. Data:
 - 87 octane: $n_1 = 6$: 26.4, 27.6, 29.7, 28.9, 29.3, 28.8
 - 90 octane: $n_2 = 6$: 30.5, 30.9, 29.2, 31.7, 32.8, 29.3
- Treat the two octane groups as **independent samples** (Welch two-sample t CI) and compute the CI for $(\mu_{90} - \mu_{87})$
- The critical value of Welch's t is 2.228 (from the Welch's t equation)

$$SE = \sqrt{\frac{s_{87}^2}{n_{87}} + \frac{s_{90}^2}{n_{90}}} = \sqrt{\frac{1.23^2}{6} + \frac{1.39^2}{6}} = 0.758$$

$$ME = t^* \cdot SE = 2.233 \times 0.758 = 1.692$$

$$(\bar{x}_{90} - \bar{x}_{87}) \pm t^* \cdot SE$$

The 95% confidence intervals $2.283 \pm 1.692 = (0.59, 3.98)$



Questions



Answers

